MILITARY SPECIFICATION SHEET

CHAFF ROLL (TYPE RR-172/ALE)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the chaff roll described herein shall consist of this document and the latest issue of Specification MIL-C-85001.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, manufacture, and acceptance requirements for a chaff countermeasures designated RR-172/ALE. This equipment consists of a chaff roll loaded with flat aluminum dipoles capable of producing confusing signals to radar systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-85001	Chaff Roll (Type RR-( )/ALE), General Specification for

2.1.1 Availability of Documents. When requesting specifications, standards and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, CODE 105, 5801 Tabor Avenue, Philadelphia, PA 19120.

3. REQUIREMENTS

3.1 General Requirements. The RR-172/ALE chaff roll shall comply with the requirements of MIL-C-85001 and this specification.

3.1.1 Item Definition. The chaff roll covered by this specification shall consist of a chaff roll assembly loaded with aluminum foil dipoles, the quantity and type described herein. This chaff roll shall be functionally compatible with the AN/ALE-41 Dispensing Set.

3.2 Performance Characteristics. Performance characteristics shall conform to the requirements of MIL-C-85001.

3.3 Detail Requirements.

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3.3.1 Chaff Roll. The RR-172/ALE chaff roll shall consist of the following:

3.3.1.1 Roll Assembly. Each chaff roll shall be contained on the roll assembly detailed in MIL-C-85001.

3.3.1.2 Chaff Roll Film. The chaff elements shall be placed between two layers of polyethylene terephthalate film (detailed in MIL-C-85001) to form a film/chaff/film sandwich.

3.3.1.3 Chaff Dipoles. The lengths and counts of chaff dipoles included within each chaff roll shall be as listed in Table I.

DIPOLE LENGTH (inches)	CUT LENGTH (inches)	TOTAL NUMBER IN ROLL
1.88 ± .06	1.88 ± .03	31 X 10^6 ± 10%
1.00 ± .06	1.00 ± .03	93 X 10^6 ± 10%
0.62 ± .06	0.62 ± .03	155 X 16^6 ± 10%
1.62 ± .06	1.62 ± .03	62 X 10^6 ± 10%

TABLE I. CHAFF DIPOLE LENGTHS AND COUNTS

3.3.1.3.1 Dipole Material. The dipole material shall be plain finish full hard aluminum foil of 1145 alloy. The foil thickness shall be 0.00035 inch ± 10%.

3.3.1.3.2 Dipole Width. Each dipole shall be flat and shall be .004 inch ±10% wide.

3.3.1.3.3 Workmanship and Finish. Workmanship and finish shall be in accordance with the requirements of MIL-C-85001.

3.3.1.4 Coating. A slip coating shall be applied to both surfaces of the dipoles. The slip coating shall be composed of the following ingredients, or equivalent, as approved by the procuring activity:

a. 18 fluid ounces of Silicone SM-2016, General Electric Company, or Silicone Syl-Off 22, Dow Corning Corporation, 40 percent aqueous emulsion.

b. 18 fluid ounces of Silicone SM-2010, General Electric Company, or Silicone DC-24, Dow Corning Corporation, 35 percent aqueous emulsion.

c. 3-1/2 fluid ounces of aluminum acetate, 20 percent aqueous solution, sulfate free.

d. 7 gallons of filtered water.

The ingredients shall be mixed at the time of use, and the reaction shall take place on the foil at the time of application. Modification of formula may be necessary to adapt to application methods.

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3.4 Assembly.

3.4.1 Film/Chaff/Film Sandwich. Each film/chaff/film sandwich shall be fabricated by laying the various chaff cuts in rows along the polyethylene film as illustrated in Figure 1. The rows shall be parallel to the leading edge of the chaff roll, and the individual chaff elements shall be perpendicular to the leading edge of the chaff roll. The 1.88 inch dipoles shall be placed in the row nearest the leading edge of the film, so that when the film/chaff/film sandwich is wound onto the hub, these dipoles shall be at the leading edge of the roll.

3.4.2 Chaff Roll. The chaff roll shall be fabricated in accordance with the requirements of MIL-C-85001.

3.4.3 Chaff Roll Weight. The minimum gross weight of the film, hub, and chaff combination shall be 46 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Inspection and testing of the RR-172/ALE chaff roll shall be conducted in accordance with the requirements of MIL-C-85001.

5. PREPARATION FOR DELIVERY

5.1 General. The chaff roll shall be prepared for delivery in accordance with the requirements of MIL-C-85001.

5.1.1 Item Description. The item description shall be "CHAFF ROLL, RR-172/ALE".

6. NOTES

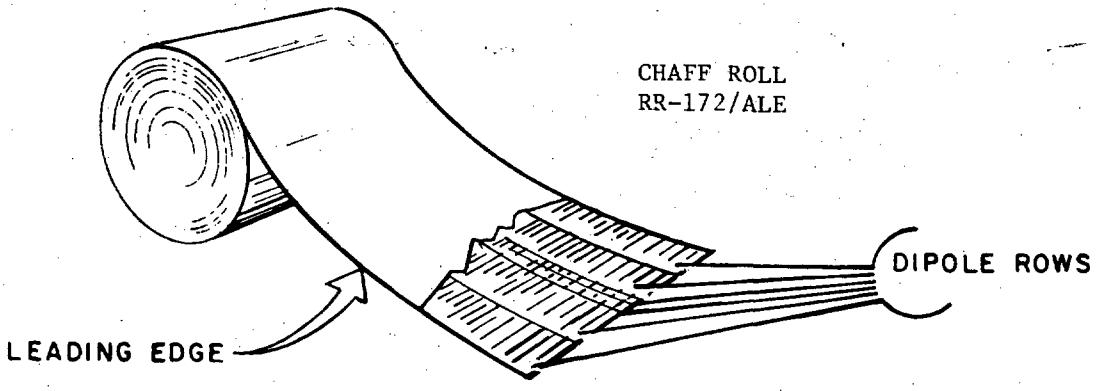
6.1 Notes. The notes contained in MIL-C-85001 shall apply.

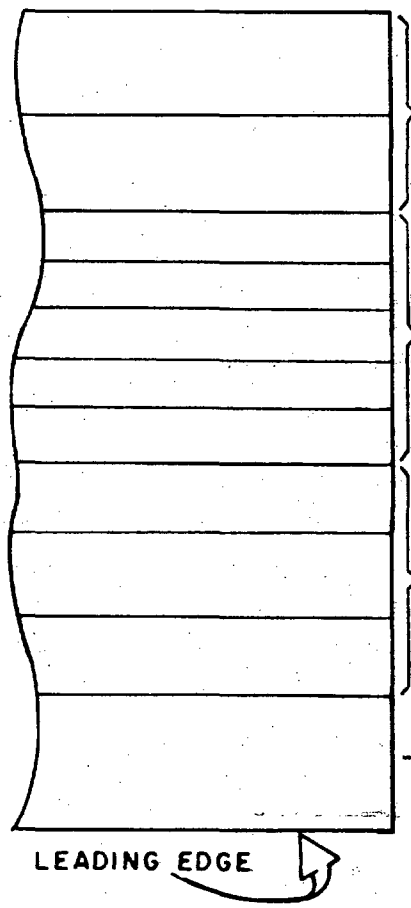
Preparing Activity:

Navy - AS

Project No. 5865-N009-2

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DIPOLE LENGTH (INCHES)	NUMBER OF ROWS
1.62	2
0.62	5
1.00	3
1.88	1

FIGURE 1

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-85001/172(AS) CHAFF ROLL (TYPE RR-172/ALE)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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